

Analysis of Variance on a Linear Model

The linear model is

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$= [1, X]\beta + \varepsilon$$

where $\beta = [\beta_0, \dots, \beta_k]$.

In ANOVA, we assume that $\varepsilon \sim N(\mu, \sigma)$. We also assume the X are independent and identically distributed.

The solution matrix for this linear model is as follows:

$$H := [1, X]([1, X]^T [1, X])^{-1} [1, X]$$

The dimension of H is N . H is idempotent, so $\text{rank}(H) = \text{trace}(H)$.

The Null and Alternative Hypotheses

Let's state the null and alternative hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

Rank of H in each model.

Let H^0 be H in the model of the null hypothesis. Let H^a be H in the model of the alternative hypothesis.

Under the null hypothesis, the linear model degenerates into

$$Y = \beta_0 + \varepsilon$$

The fit of this model is given by $\beta_0 = \bar{Y}$.

The form of H^0 is J/N , where J is a matrix of all ones.

The rows of H^0 are identical, so the rank of H^0 is 1.

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Under the alternative model, we get the full predictor, $\hat{Y}$:

$$\hat{Y} = \beta_0 + \beta_1 X$$

Because we assume i.i.d. variables, the N X are independent. The rank of H^a is $\text{rank}([1, X]) = 2$

Degrees of Freedom of Residuals and Regressions

Three residual distributions are considered:

$$SSR = \|\hat{Y} - \bar{Y}\|^2$$

$$SSE = \|Y - \hat{Y}\|^2$$

$$SST = \|Y - \bar{Y}\|^2$$

SST is the sum of squared residuals of the null model. Expressed in terms of the solution matrix, H^0 :

$$Y - \bar{Y} = (I - H^0)Y$$

I , H^0 , and $I - H^0$ are idempotent, so we have

$$\begin{aligned} \text{rank}(I - H^0) &= \text{tr}(I) - \text{tr}(H^0) \\ &= \text{rank}(I) - \text{rank}(H^0) \\ &= N - 1 \end{aligned}$$

The degrees of freedom of SST is $N-1$.

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SSE is the sum of squared residuals of the alternative model. Expressed in terms of the solution matrix:

$$Y - \hat{Y} = (I - H^a)Y$$

I , H , and $I - H^a$ are all idempotent, so we have

$$\begin{aligned} \text{rank}(I - H^a) &= \text{tr}(I) - \text{tr}(H^a) \\ &= \text{rank}(I) - \text{rank}(H^a) \\ &= N - 2 \end{aligned}$$

The degrees of freedom of SSE is $N - 2$.

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SSR is the regression of $\hat{Y}_i$ onto $\bar{Y}$. Expressed in terms of a solution matrix:

$$\begin{aligned} \hat{Y}_i - \bar{Y} &= (H^a - H^0)Y \\ &= (H^a + (I - I) - H^0)Y \\ &= ((I + H^a) - I - H^0)Y \end{aligned}$$

H^a , H^0 , and $(I - H^0)$ are symmetric and idempotent.

$$\begin{aligned} \text{rank}((I + H^a) - I - H^0) &= \text{tr}(I) + \text{tr}(H^a) - \text{rank}(I - H^0) \\ &= N + \text{rank}(H^a) - N - 1 \\ &= N + 2 - N - 1 \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

The degrees of freedom of SSR is 1.

SSR and SSE may be Chi-Squared

Let's restate Cochran's Theorem.

If N observations, Y , come from the same $N(\mu, \sigma^2)$, and SST is decomposed into k SS_r , with degrees of freedom, df_r , then each $SS_r/\sigma^2 \sim \chi^2(df_r)$ if $\sum_r^k df_r = N - 1$.

Under the null hypothesis, we have $Y \sim N(\mu, \sigma^2)$ as a consequence of $\varepsilon \sim N(\mu, \sigma^2)$. SST is decomposed into SSR and SSE , such that $df_r + df_e = N - 1$.

We conclude that, if the null hypothesis is not discharged, then $SSE \sim \chi^2(N - 2)$ and $SSR \sim \chi^2(1)$.

The ratio of MSR and MSE may form an F-distribution

Recall that $MSR = SSR/df_r = SSR$, and $MSE = SSE/df_e = SSE/N - 2$.

If SSR and SSE are χ^2 -distributions, the ratio of $SSR/df_r/SSE/df_e$ is an F-distribution.